Lists and Tables

HTML Lists

HTML lists are used to organise and structure content on a web page. Lists allow you to group related items together, making the information more readable and organised for users. HTML provides three main types of lists:

a. Ordered Lists
b. Unordered Lists
c. Definition Lists

Alt text: *A Web Page showing use of tables and lists*

a. Ordered Lists (‹ol›):

- Ordered lists are used when the order of the items is important.
- Each item in an ordered list is marked with a number (by default, it's a decimal number, but you can change it to Roman numerals or letters using the type attribute).
- The ‹ol› element is used to create an ordered list, and each list item is defined with the ‹li› (list item) element.

‹ol›- Used to create an ordered list in HTML. Common values of type attribute include "1" (decimal, default), "A" (uppercase letters), "a" (lowercase letters), "I" (uppercase Roman numerals), and "i" (lowercase Roman numerals).

Attribute value	Output
1	decimal, default
A	uppercase letters
a	lowercase letters
I	uppercase Roman numerals
i	lowercase Roman numerals

<li>- is the list item, which defines individual items within the list.

Example:

HTML Code	Output

`<body>`
`  <ol type="A">`
`  <li>Books</li>`
`  <li>Pens</li>`
`  <li>Pencil</li>`
`  <li>Eraser</li>`
`  </ol>`
`</body>`
	A. Books
	B. Pens
	C. Pencil
	D. Eraser

This HTML code creates an ordered list (<ul>) with a numbering style with uppercase alphabets . Inside the list, there are four list items (<li>), each representing a different item: "Books," "Pens," "Pencil," and "Eraser."

b. Unordered Lists (<ul>):

- Unordered lists are used when the order of the items is not important, and the items are typically marked with bullets or other symbols.
- The <ul> element is used to create an unordered list, and each list item is defined with the <li> element.

<ul>: This tag is used to create an unordered list in HTML. Type attribute can be used with <ul> tag to change the bullets. Common values of type can be disc, circle, square, disc.

Attribute value	Output
disc	●
circle	○
square	■

Example:

HTML Code	Output

`<body>`
`  <ul type="circle">`
`  <li>Books</li>`
`  <li>Pens</li>`
`  <li>Pencil</li>`
`  <li>Eraser</li>`
`  </ul>`
`</body>`
	○ Books
	○ Pens
	○ Pencil
	○ Eraser

The above HTML code creates an unordered list (<ul>) with a circular bullet style (type="circle"). Inside the list, there are four list items (<li>), each representing a different item: "Books," "Pens," "Pencil," and "Eraser."

c. Definition Lists (<dl>):

- Definition lists are used to define terms and provide corresponding descriptions.
- The **<dl>** element is used to create a definition list, and terms are marked with the **<dt>** (definition term) element, while the corresponding definitions are marked with the **<dd>** (definition description) element.

HTML Code	Output

`<dl>`
`  <dt>Term 1</dt>`
`  <dd>Definition 1</dd>`
`  <dt>Term 2</dt>`
`  <dd>Definition 2</dd>`
`</dl>`
	Term 1
	Definition 1
	Term 2
	Definition 2

- <dl>: This is the definition list container.
- <dt>: Stands for "definition term." It marks the start of each term.
- <dd>: Stands for "definition description." It provides the description or definition for each term.

HTML Tables

If you need to arrange data in rows and columns you need to create a table. Following tags are used to create a table:

Tag	Description
<table>	Creates a table in web page
<tr>	Stands for Table row, creates a Row in the table
<th>	Stands for Table header, creates a header in table
<td>	Stands for table data, creates table cell
<colspan>	Stands for column span , Single cell span over multiple columns
<rowspan>	Stands for row span, Single cell span over multiple row

Attributes of table-

The <table> tag in HTML has various attributes that allow you to customise and structure the table. Here are some common attributes associated with the <table> tag:

a. **border:** Specifies the width of the border around the table.
   <table border="value">
b. **Width:** Sets the width of the table.
   <table width="value">
c. **cellpadding:** Specifies the space between the content of a cell and its border.
   <table cellpadding="value">
d. **cellspacing:** Defines the space between cells in the table.
   <table cellspacing="value">
e. **summary:** Provides a summary or description of the table for accessibility.
   <table summary="text">
f. **caption:** Allows the insertion of a caption for the table, providing additional information.
   <caption>Caption text</caption>
g. **background:** Sets a background image for the entire table.
   <table background="URL">

Let us now learn the following-

a. Create a table
b. Create a Row
c. Create Table Heading
d. Create Table Data
e. Merge two Rows
f. Merge two Columns

a. Create a table- The <table></table> tag is used to create a table.

HTML Code	Output

`<body>`
`  <table border="1">`
`  </table>`
`</body>`
	This will create a table but you will not be able to see the table.

b. Create a row- The <tr> will create a row in the table.

HTML Code	Output

`<table border="1">`
`<tr>`
`</tr>`
`</table>`
	This will create a row but you will not be able to see the table.

c. Create Table Heading- To make the row visible we have to add some data. So we will make the first row of the table as heading. <th> will create headings of the table as Name, Email, Phone.

HTML Code	Output

`<table border="1">`
`<tr>`
`<th>Name</th>`
`<th >Email</th>`
`<th>Phone</th>`
`</tr>`
`</table>`
	Name Email Phone

This HTML code creates a table with a border attribute set to "1", indicating the presence of table borders. The table structure includes a table row (<tr>) containing three table header cells (<th>): "Name," "Email," and "Phone."

d. Create Table Data- The <table></table> tag is used to create tables. <tr> will create a row in which email, name, phone will be heading to the first row.

HTML Code	Output

`<table border="1">`
`<tr>`
`<th>Name</th>`
`<th >Email</th>`
`<th>Phone</th>`
`</tr>`
`<tr>`
`<td >John Doe</td>`
`<td>john@example.com</td>`
`<td>Additional Email</td>`
`<td>555-1234</td>`
`</tr>`
`</table>`
	Name Email Phone
	John Doe john@example.com Additional Email 555-1234

Above HTML code creates a simple table with a border attribute set to "1," indicating the presence of table borders. The table consists of two rows (<tr>). The first row contains three table header cells (<th>),"Name," "Email," and "Phone." The second row has four table data cells (<td>), "John Doe," "john@example. com," "Additional Email," and "555-1234."

e. **Merge two Rows:** You can use the rowspan attribute to merge two columns. rowspan is helpful when you need to merge cells vertically**.**

HTML Code	Output

`<body>`
`  <table border="1">`
`  <tr>`
`  <th>Name</th>`
`  <th>Email</th>`
`  <th>Phone</th>`
`  </tr>`
`  <tr>`
`  <td rowspan="2">John Doe</td>`
`  <td>john@example.com</td>`
`  <td>Additional Email</td>`
`  <td>555-1234</td>`
`  </tr>`
`  <tr>`
`  <!-- This cell is empty as the 'Name' cell above spans two rows -->`
`  <td>john@example.com</td>`
`  <td>555-5678</td>`
`  </tr>`
`  </table>`
`</body>`
	<table><tr><th>Name</th><th>Email</th><th>Phone</th><tr><td rowspan="2">John Doe</td><td>john@example.com</td><td>Additional Email</td><td>555-1234</td><tr><td>john@example.com</td><td>555-5678</td></table>

This HTML code represents how to use rowspan. The second row contains the first table data cell (<td>) with rowspan="2", indicating that the "Name" cell spans two rows.
The second and third cells contain email addresses. The fourth cell displays a phone number.
In the second row, the "Name" cell's rowspan causes it to cover two rows, creating a visually distinct table structure.

f. Merge two Columns: You can use the colspan attribute to merge two columns. colspan is helpful when you need to merge cells horizontally.

HTML Code

```
<body>
    <table border="1">
        <tr>
          <th>Name</th>
          <th
colspan="2">Email</th>
          <th>Phone</th>
        </tr>
        <tr>
          <td >John Doe</td>
          <td>john@example.
com</td>
          <td>Additional
Email</td>
          <td>555-1234</td>
        </tr>
        <tr>
          <td >Ritu
Goyal</td>
          <td>john@example.
com</td>
          <td>555-5678</td>
        </tr>
      </table>
</body>
```

Output

Name	Email		Phone
John Doe	john@example.com	Additional Email	555-1234
Ritu Goyal	john@example.com	555-5678	

In the above code the second row contains table data cells (<td>) with information for "John Doe," including two cells for email ("john@example.com" and "Additional Email") and one for the phone number ("555-1234"). The third row provides information for "Ritu Goyal," with corresponding data in the "Name," "Email," and "Phone" columns.